

Lead AI and ML Engineer · LLM Systems & Production GenAI · GCP · Azure · GPU Infra

Lead AI and ML engineer and data scientist with 10+ years of engineering experience across classical ML, deep learning, computer vision, NLP, forecasting, operations research, and LLM fine-tuning and serving. Design, deploy, scale, and maintain high-throughput, low-latency ML systems-from training and evaluation to inference, monitoring, and retraining. Lead architecture, mentoring, and cross-functional delivery. Delivered **\$300K/year** LLM savings; fraud scoring at **40,000+ events/sec** under a **200 ms SLA**; AWS security-model training on **billion-row datasets**; and distributed computer-vision training on **~1M images in ~3 hours**.

Skills

- **LLM Training & Serving:** LLM Fine-tuning; LoRA / QLoRA; LLM Evaluation; OpenAI-Compatible Serving; Multi-Task Adapters
- **Modelling & Analytical Methods:** Classical ML; Deep Learning; Computer Vision; Natural Language Processing (NLP); Time-Series Forecasting; Feature Engineering; Behavioral Anomaly Detection; Autoencoders; Operations Research; Vehicle Routing; Mixed-Integer Programming; Constraint Modelling; Supply-Chain Simulation
- **Frameworks & APIs:** FastAPI; scikit-learn; XGBoost; PyTorch; TensorFlow / Keras; TensorFlow Serving; Hugging Face Transformers / PEFT; Unsloth; vLLM; Google OR-Tools
- **GPU & Distributed Computing:** Distributed Multi-GPU Training; CUDA; GPU-Accelerated Computing; RAPIDS cuDF / CuPy
- **Experimentation & Model Lifecycle:** MLflow; Experiment Tracking; Model Evaluation; Model Versioning / Registry; Automated Retraining; Vertex AI Pipelines; Azure ML Pipelines; Vertex AI Feature Store
- **Cloud & Serverless Services:** GCP; Microsoft Azure; GKE; Vertex AI; Azure ML; Cloud Run; Cloud Functions; Azure Function Apps; Logic Apps
- **Deployment & Production Reliability:** Docker; Kubernetes; Helm; Seldon Core; Istio; HPA Autoscaling; CI/CD; Cloud Build; Azure DevOps / Pipelines; ArgoCD; Google KCC; GitOps; JWT Authentication
- **Monitoring, Testing & Reporting:** Prometheus; Grafana; Google Cloud Monitoring; Model Monitoring; Load Testing; Streamlit; Power BI
- **Languages, Data & Storage:** Python; SQL; Apache Beam / Dataflow; Pub/Sub; BigQuery; MS SQL Server; Redis Caching; GCS; Azure Blob Storage; Artifact Registry; Pandas / NumPy; Dask; Git; Linux / Shell Scripting

Experience

Lead Artificial Intelligence Engineer

Best Buy | Jul 2025 - Present

- **LLM Fine-tuning Framework:** Built config-driven LoRA/QLoRA fine-tuning with Unsloth and PEFT. Supported single/multi-GPU runs, multi-task adapters, and automated evaluation. YAML task configuration and standard PEFT artifacts enabled reusable tasks sharing one vLLM deployment.
- **LLM Serving & Impact:** Deployed OpenAI-compatible vLLM endpoints on GKE with HPA and Istio; integrated GCS, Artifact Registry, Cloud Build, and Prometheus/Grafana. Self-hosting eliminated paid API dependency, saved **\$300K/year**, and strengthened data governance and compliance posture.
- **Retail Fraud Detection & Prevention:** Built Apache Beam/Dataflow streaming and GKE inference for XGBoost fraud models at **40,000+ events/sec** under a **200 ms SLA**. Improved fraud-score coverage by **~20%** during peak holiday traffic. Integrated Vertex AI feature, registry, and training services, secure endpoints, load testing, monitoring, and GitOps across payment, refund, and reship signals.
- **Leadership & Delivery:** Lead architecture and production delivery across search, personalization, fraud, and cybersecurity; partner with data scientists to turn validated models into secure, monitored services.

Lead ML Engineer - ML Engineering & Operations

Wipro | Aug 2023 - Jul 2025

- **AWS Account Digital Fingerprinting:** Engineered GPU-accelerated training and inference with RAPIDS cuDF/CuPy on NVIDIA DGX (8xA100). Trained per-entity behavioral autoencoders on **billion-row AWS security datasets** spanning 20+ days, learning normal activity fingerprints and surfacing real-time anomalies. Integrated continuous retraining, MLflow model management, and Grafana.
- **Proofpoint Phishing Detection:** Built training and inference pipelines scoring **2,400 emails/sec**, double the previous rate. Used security-team feedback and Microsoft Sentinel integration to detect phishing missed by initial filters and accelerate quarantine response.
- **Brand Infringement & Abuse-Inbox Analysis:** Built ML routing and OCR extraction of scammer identifiers across text, HTML, images, and attachments; routed evidence to security teams and mail partners for takedowns. Automated triage **doubled the prior case volume** and reduced manual review.

- **Leadership & Delivery:** Led and mentored ML/MLOps engineers; coordinated client/NVIDIA delivery and scaled the India AI hub through **300+ résumé screens** and **50+ technical interviews**.

Senior Data Scientist - Data Science / Azure ML

Accenture | May 2021 - Sep 2023

- **Product Quality Prediction - BamaGruppen:** Built Azure ML warehouse-arrival quality prediction for the Norwegian distributor, with automated retraining, Logic Apps, Azure Pipelines, storage, and stakeholder alerts. Increased truck inspections from **130 to 160/day**, defect detection by **17%**, and customer approval by **20%**, delivering **\$2.5M annual savings**.
- **Last-Mile Vehicle Routing:** Designed OR-Tools vehicle-routing models with capacity, delivery-window, waiting-time, and store-demand constraints. Compared optimized routes with manual plans to improve resource utilization and reduce logistics costs.
- **Mellom Transport Optimization:** Built mixed-integer models for daily terminal-to-distribution-center transport, with Azure Function Apps APIs, Logic Apps, RPA, CI/CD, and Slack alerts. Replaced manual Excel planning with optimized, automated, scalable plans and rapid adaptation to changing operational constraints.
- **Norsk Oppstrom - Simulation Framework:** Modelled product flows across vendors, warehouses, hubs, and customers, comparing cost, freshness, and carbon-footprint scenarios. Integrated simulation results with Power BI and identified the appropriate number and locations of regional hubs.
- **Hub Analytics POC:** Simulated strategic distribution hubs to identify beneficiary customers/vendors and assess warehouse-volume reduction.
- **Leadership & Delivery:** Led cross-functional delivery, mentored data scientists, and presented findings and recommendations to Norwegian stakeholders.

Data Scientist

GD Research Centre Pvt Ltd | May 2020 - May 2021

- **GlobalData Forecasting:** Built domain-aware forecasts and scenario analysis across **15+ industry verticals**, informing strategic market-intelligence reports and resource-allocation decisions.

Deep Learning Engineer

Tata Consultancy Services | Jul 2019 - Mar 2020

- **Computer Vision - Semiconductor Defect Classification:** Built distributed multi-GPU DenseNet training on **~1 million images in ~3 hours**, achieving **90% accuracy** and **70% precision/recall on rare classes**. Led inference/data-platform development, deployed Docker/TensorFlow Serving at Amkor in South Korea, and trained the client team to operate the system.

Senior Software Engineer - ML

Infosys | May 2016 - Jul 2019

- **Opportunity Win/Lose Prediction:** Built logistic-regression scoring with SSIS data pipelines and Power BI/SSAS dashboards for data-driven sales prioritization.
- **Automated ID-Card Generation:** Delivered tested new-member insurance workflows, from impact analysis and design through implementation, support, and team knowledge sharing.

Open-Source Project

vLLM Model Chooser

Built an interactive GPU-to-model matchmaker with VRAM, quantization, and KV-cache fit filters. [Demo](#).

Certifications

- Google Cloud Certified: Generative AI Leader | Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure Data Scientist Associate | Microsoft Certified: Power BI Data Analyst Associate
- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional

Education

B.E. Electrical & Electronics

Andhra University, Visakhapatnam | 2012 - 2016 · 80%